Pranav Kumar Soma

MACHINE LEARNING ENGINEER • AI RESEARCH

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EDUCATION

M.S. Computer Science, UC San Diego Dec 2026 (expected)

GPA 4.00 • Jacobs School of Engineering • M.S. in 1.25 years

B.S. Computer Science, UC San Diego Dec 2025

GPA 3.96 • Cum Laude, CSE Department Honors, Provost Honors (all quarters) • B.S. in 2 years

EXPERIENCE

Founding Software Engineer | Stealth Startup Starting Sep 2026

Autonomy & perception • part-time

- Building **perception & autonomy**: multi-sensor **data-fusion** pipelines, real-time object tracking, and **multi-agent** trajectory planning & resource coordination across distributed nodes, validated in simulation and **hardware-in-the-loop** testing

AI Research Fellow & Agentic Systems Lead | Laboratory for Emerging Intelligence, UCSD Apr 2025–Present

- Fine-tuned model-agnostic tutoring LLMs (**LoRA + SFT + RLHF**) with **Bayesian Knowledge Tracing**; deployed AI Tutor to 3 universities for **8,000 students / 8 courses** · **64% fewer TA hours, 87% higher engagement** (A/B tested across assignments)
- Lead **Teradata-funded** agentic-AI research · custom **MCP servers** over UCSD APIs, automated skill generation & optimization, interpretability of user intent vs. skill instructions; MongoDB vector store, FastAPI serving

Software Engineer Intern | ServiceNow Summers 2025 & 2026

*AI Voice Agents • **Technical Skills Award** · top intern of 800+*

- Cut **AI Voice Agent latency 45%** by re-architecting the **NLU** pipeline onto higher-capacity base models across 8 intents; shipped to production and validated live with **Disney+**. **Return offer; invited back for a second summer**

Graduate Researcher | Qualcomm Institute / Calit2 (HXI Lab) Nov 2024–Dec 2025

- Built **multimodal** ML pipelines over **TB-scale** sensor/audio/video/image data, parsing patient signals into structured health-database queries for the LifeSaver AI assistant (emergency services, elderly, rural care)
- Fine-tuned locally hosted **Qwen** models (SFT) on the SD Supercomputer Center; built **GraphRAG** over the UMLS medical ontology · accuracy validated against **EMS triage protocols** at sub-second latency; engineered the robotics pipeline tying queries to servo actuation. **First-author** lab poster

Honors & M.S. Thesis Researcher | UCSD CSE · Computational Redistricting Sep 2024–Present

- Trained per-target **XGBoost** regressors on per-block geospatial/demographic features + **multilevel graph partitioning** to generate districting plans (pooled fav-share **MAE 0.042** across 9 states, held-out cross-state generalization)
- Implemented a **Sequential Monte Carlo** sampler (spanning-tree balanced cuts, N=128 particles) and greedy DP baselines; authored VRA Section-2 / Polsby–Popper / efficiency-gap metrics; reproduces official district counts for AL, CA, TX

SELECTED PROJECTS

- **ORCA · AI Music Production** *(Best Project, SDTC: 1 of 30 from 200+; SAIRS 2025)* · hybrid neural networks for source separation, CNN audio-to-score, OpenCV vision; **98% arrangement accuracy** (SME, layman & cross-entropy validated)
- **BioVeritas** · AI platform evaluating logical rigor & authenticity in biological research via retrieval/context systems

LEADERSHIP

Co-Founder & President, Real World AI Network *(2026–Present)* · founded **7,000+ student** federation across UCSD's engineering, data-science & business schools driving AI research & entrepreneurship.

Founding Board Member & VP External Affairs, University AI Alliance · student-led frontier-tech network spanning MIT, Stanford, Caltech & Cornell; own external relationships with a16z speedrun, Bain Capital Ventures, Techstars & execs at Google and Meta.

President, CSE Society @ UCSD (200+ members, 21+ projects) • **Founder**, Innovate Research Group (30+ across UCLA, UCSB, UCSC) • **Awards:** National Merit Finalist, USACO Silver, AP Scholar w/ Distinction, YC Startup School 2026.

TECHNICAL SKILLS

ML & AI: PyTorch, Transformers, scikit-learn, XGBoost, OpenCV; LLM fine-tuning (LoRA, SFT, RLHF), RAG & GraphRAG, agents & MCP, LLM-as-judge evals, multimodal models, CNNs/RNNs/LSTMs, RL, NLP/NLU, Bayesian Knowledge Tracing, Sequential Monte Carlo, sensor fusion, computer vision; Ollama, Qwen

Languages: Python, Java, C++, C, JavaScript, TypeScript, SQL, Swift, Kotlin, ARM-32 Assembly, Bash, HTML/CSS

Data & HPC: NumPy, Pandas, GeoPandas, Shapely, pymetis, Parquet/PyArrow, multi-GPU & parallel computing, SD Supercomputer Center, spatial databases

Infra & Tooling: FastAPI, Flask, React, Node/Express; MongoDB (vector), PostgreSQL; Docker, AWS (Lex, Lambda), Git, Linux, Jupyter, ROS, embedded & distributed systems